



Review article

Sustainable synthesis of silica nanoparticles (SNPs) from agricultural residues: A review on plant stress mitigation and the sustainability triangle

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ABSTRACT

Silica nanoparticles (SNPs) have shown potential in a variety of applications across industries, including agriculture. Amid growing concerns about sustainable agricultural intensification and the environmental burden of synthetic agrochemicals, there is a pressing need for eco-friendly alternatives to support crop resilience. These nanoparticles can be sustainably extracted from agricultural residues such as rice hulls/husks, sugarcane bagasse, and corn cobs. These by-products are not only rich in silica but are also produced in large quantities annually, making them ideal candidates for SNPs production. This review discusses the synthesis methods of SNPs from agricultural residues and their potential to enhance crop productivity, particularly under abiotic stressors. The novelty of this work lies in its focus on the sustainability triangle – economic, environmental, and social sustainability – in the context of SNPs production from agricultural residues. Studies suggest that SNPs can mitigate the negative effects of various abiotic stresses, including salinity, drought, and heavy metal toxicity. However, despite their promising potential, the SNPs extraction process needs further optimization, considering the different isolation methods and process variables. To ensure scalable and responsible implementation, future research should also incorporate techno-economic analysis (TEA), life-cycle assessments (LCA), and relevant social sustainability factors.

1. Introduction

The global population continues to grow at an unprecedented rate, necessitating a parallel increase in food production and raw materials from diverse agricultural crops. A meta-analysis of global food demand and hunger risk projections for 2010–2050 indicates that total food demand is expected to rise by 35% to 56%, while the population at risk of hunger may change by –91% to +8% during the same period [1]. To meet this ever-increasing demand, innovative approaches are required to enhance crop productivity and sustainability. One way to achieve this goal is by improving agricultural resilience to environmental challenges, ensuring that crop yields remain stable despite adverse growing conditions. As global food security becomes an urgent priority, there is an escalating demand for technologies and strategies that safeguard agricultural systems from potential threats and maximize their productivity.

However, crops face numerous challenges from diverse abiotic stresses, including salt stress, drought, and heavy metal contamination

[2,3]. These stresses can severely impede plant growth, leading to reduced crop productivity, with crop yield loss ranging from 50% to 70% [4,5]. In response to these stresses, plants often exhibit cellular dehydration, stomatal closure, osmotic imbalance, disruption of photosynthesis, accumulation of reactive oxygen species (ROS), inhibition of enzymatic defense mechanisms, and alterations in gene expression, ultimately resulting in reduced growth rates and diminished nutritional quality [6,7]. Plants cultivated in saline soils and polluted environments often absorb excessive sodium ions and heavy metals, leading to stunted growth, impaired development, and decreased yield [8]. Similarly, water scarcity and drought stress can disrupt cellular homeostasis, leading to reduced turgor pressure, impaired nutrient uptake, and decreased photosynthetic activity [9]. These conditions often trigger oxidative stress, membrane damage, and metabolic imbalances, further exacerbating growth inhibition, developmental delays, and significant yield losses.

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In response to these abiotic stressors, numerous proposed and applied solutions have emerged to mitigate their adverse effects and improve crop resilience. Conventional approaches include the development of stress-tolerant crop varieties through breeding programs [10–12], optimized irrigation systems [13,14], soil amendments (such as vermicompost, vermiwash, biochar, bio-fertilizers, and plant growth-promoting rhizobacteria) [15], and the application foliar fertilizers [16] and organic and inorganic fertilizers to enhance soil fertility and plant health. Additionally, biotechnological strategies, such as genetic engineering and the use of plant growth-promoting rhizobacteria (PGPR), have shown promise in improving stress tolerance by enhancing antioxidant defense systems and osmoprotectant synthesis [17,18]. More recently, advancements in nanobiotechnology have opened new avenues for enhancing plant performance under stress conditions [19–23]. By incorporating nano-fertilizers, nano-elicitors, and nano-pesticides, nanotechnology presents an innovative strategy to promote plant growth, enhance productivity, and increase resilience to stress [24,25]. Engineered nanoparticles, including silica nanoparticles (SNPs), have gained attention due to their ability to improve nutrient availability, strengthen plant cell walls, and mitigate oxidative damage, thus enhancing stress tolerance and overall crop productivity [23].

SNPs are inorganic nanomaterials primarily synthesized from silica precursors, such as sodium silicate, tetraethyl orthosilicate (TEOS), and rice husk ash, through sol-gel, precipitation, and hydrothermal methods [26,27]. They possess a high surface area, tunable pore size, and excellent thermal and chemical stability, making it suitable for diverse applications (Table 1), including in agriculture. Chemically, SNPs consist of silicon dioxide (SiO_2) arranged in a tetrahedral structure, where each silicon atom is bonded to four oxygen atoms, forming a large matrix of silica [28,29]. These tetrahedral units can create pores ranging in size from 5 to 3000 Å and can exist in either amorphous or crystalline forms, such as quartz. The bond angles between silicon-oxygen-silicon bonds, known as siloxane bonds, range from 120° to 180°, influencing the material's structure and properties [30]. The surface of SNPs typically contains silanol groups (Si-OH), which can be functionalized to enhance interactions with biomolecules [31]. These characteristics allow SNPs to act as a carrier for nutrients, agrochemicals, and bioactive compounds, enabling controlled release and improved bioavailability [32]. Additionally, SNPs exhibit antioxidant properties by scavenging ROS and reinforcing plant cell walls, thereby enhancing stress tolerance and promoting growth under adverse conditions such as salinity, drought, and heavy metal toxicity.

SNPs have not only gained significant attention due to their unique properties but also because they can be sustainably extracted from diverse agricultural residues, contributing to waste valorization and promoting a circular economy [82,83]. Agricultural by-products such as rice husk [30,84,85], wheat straw [86–88], sugarcane bagasse [89,90], and corn stalk [91,92] are rich in silica and provide an environmentally friendly and cost-effective source of SNPs. The ability to extract silica from these residues aligns with global sustainability goals outlined by the United Nations, particularly in relation to the Sustainable Development Goals (SDGs), such as responsible consumption and production (SDG 12) and climate action (SDG 13). Utilizing agricultural waste for nanomaterial production helps reduce environmental pollution and enhances resource efficiency, contributing to a more sustainable agricultural system [82,83,93,94].

With the promising applications of SNPs, numerous reviews have emerged on the synthesis methods of silica and SNPs from a wide range of sources, including solid wastes from agricultural, industrial, mining, municipal sectors and geothermal sludge [19,20,95–98]. In terms of its multidisciplinary applications, reviews emerged on the bibliographic analysis and comprehensive review on the application of SNPs in concrete pavement [99], cement-based materials [100–104], applications for high-performance lithium-ion battery [105], in biomedical applications such as drug delivery systems [106–108], and for packaging and coating applications [103,104]. Moreover, SNPs has

emerged as a promising approach for mitigating abiotic stress in agricultural crops. Recent studies have shown that SNPs improved the salt tolerance of diverse agricultural crops such as banana [109], pea plant [110], faba bean [111], grapes [112] and rice [112]. SNPs have also been widely used in improving the drought tolerance [113–118], water stress [119,120], and heavy metal associated stresses [121–123].

Although in-depth reviews on the accumulation and mechanisms of SNPs in diverse crops are already available in the works of Grewal et al. [24,25] and Mathur and Roy [23], readers are encouraged to consult these articles for further details. This review article briefly examines the potential of agricultural residues for SNPs production. It also presents recent advances in the application of SNPs for abiotic stress tolerance in a variety of crops, including both agricultural and medicinal plants, with a specific focus on salt stress, water or drought stress, and heavy metal contamination. The novelty of this work lies in its focus on the recent progress and future trends in sustainability triangle—economic, environmental, and social sustainability—in the context of SNPs production from agricultural residues. While most studies emphasize the technical and material aspects of SNPs synthesis, this review uniquely integrates the three pillars of sustainability, providing a comprehensive approach to evaluating the potential of SNPs. By highlighting the economic feasibility, environmental impact, and social implications of SNP production, this work aims to offer a balanced perspective on the sustainability of using agricultural residues for SNP synthesis. Furthermore, this review article will discuss potential research areas in SNPs production based on the three pillars of sustainability.

2. Overview of SNPs derived from agricultural residues

Silica nanoparticles (SNPs) derived from agricultural wastes represent an attractive and sustainable precursor for silica nanoparticle production due to their low economic cost, high availability, and environmentally friendly nature. Agricultural residues, such as rice husks, sugarcane bagasse, and corn cobs, are rich sources of silica, predominantly in amorphous form, which facilitates easier extraction and processing into nanoscale materials. These agricultural residues are considered valuable sources of SNPs not only because of their silica-rich content but also due to the large volumes in which they are produced. The statistics in Fig. 1 reflect the global production of milled rice, sugarcane, and corn by country. In 2022–2023, approximately 437.75 MMT of milled rice, 1730.25 MMT of sugarcane, and 1235.73 MMT of corn were produced [124–126]. This large volume of production corresponds to a substantial amount of generated waste and by-products. Although numerous studies have been published on other agricultural residues, such as SNPs extracted from millet husk [127,128], teff straw [129,130], bamboo leaves [131], olive stones [132], and other agricultural residues [133], this section focuses on rice husk/hull, sugarcane bagasse, and corn cobs. These three residues were selected because they are the most widely researched in the past four years (see Table 2).

3. Synthesis of SNPs from agricultural residues

The extraction of silica entails the elimination of impurities or undesirable components from the raw materials (agricultural residues). This can be achieved through various techniques, including physical (thermal), chemical, and microbial processes. The physical (thermal) process involves calcination or pyrolysis process, where agricultural residues are subjected to high temperatures (550–750 °C) to remove organic matter [30,155], leaving behind silica-rich ash for the extraction of high-purity SNPs [84]. This method is widely utilized due to its simplicity and efficiency in producing amorphous silica [156]. Additionally, the extraction of SNPs through this method not only eliminates most of the organic compounds in the raw materials but also generates thermal energy and steam, which can be utilized to power turbines and produce electricity [157]. The generated product after this

Table 1
Applications silica nanoparticles (SNPs) in a variety of industries.

Field	Applications of SNPs	References
Agriculture	> Nanofertilizer for nutrient delivery > Improves soil aeration and water retention > Enhances pest resistance > Carries pesticides and herbicides > Mitigates drought and salinity stress > Boosts seed germination > Increases nutrient uptake efficiency > Reduces heavy metal toxicity > Enhances photosynthesis > Promotes root growth and microbial activity	[23,33–40]
Automotive	> Improves tire wear resistance > Enhances thermal stability in engines > Adds anti-reflective and anti-fog coatings > Increases fuel efficiency > Reinforces rubber flexibility and strength > Provides scratch resistance in paints > Supports lightweight, fuel-efficient designs > Strengthens adhesives for bonding parts > Boosts corrosion resistance in coatings	[41–52]
Biomedical	> Enables controlled drug release > Supports targeted drug delivery > Stabilizes vaccines and biologics > Provides scaffolds for tissue engineering > Enhances diagnostic imaging > Facilitates gene delivery and therapy > Improves implant biocompatibility > Offers antimicrobial coatings > Biosensors for disease detection > Promotes wound healing	[53–63]
Construction	> Improves concrete strength and durability > Enhances crack resistance and shrinkage control > Acts as a filler in cement composites > Increases water and corrosion resistance > Improve adhesion in coatings and paints > Provides UV and fire resistance > Supports self-healing concrete systems > Boosts thermal insulation	[64–73]
Other industries	> Textile Industry > Electronics Industry > Food Industry > Packaging industry > Energy Industry > Cosmetics Industry > Wastewater treatment > Aerospace Industry > Paper and Packaging Industry > Mining Industry > Paints and Coatings Industry > Electronics and semiconductor	[74–81]

process is termed as rice hull ash (RHA), sugarcane bagasse ash (SCBA), and corncob ash (CCA) which typically contains 85 wt.% [157], 53% (w/w) [158], and >60 wt% [159] silica content, respectively. It is important to note that variations in actual experiments may arise due to factors such as environmental conditions, geographical location, and soil characteristics where these crops were cultivated and grown.

It is also important to recognize, however, that the direct combustion of these residues generates substantial emissions of particulate matter, such as CO₂, CH₄, CO, NO_x, SO₂, PM_{2.5}, and PM₁₀, along with greenhouse gases [30]. These emissions not only contribute to air pollution and global warming but also pose serious health risks, including respiratory and cardiovascular diseases. To address this issue, it is essential to incorporate both dust collectors and gas absorbers when employing this method. Another approach for SNPs synthesis from agricultural residue is the chemical process which employs acids, alkalis, or solvents to dissolve impurities and extract silica from agricultural residues. This method often involves acid leaching (e.g., HCl or H₂SO₄) to remove metallic impurities and alkaline treatment (e.g., NaOH or

KOH) to dissolve lignin and amorphous silica, followed by precipitation or sol–gel techniques to produce SNPs with controlled size and morphology. The chemical process is highly effective in yielding high-purity SNPs due to its ability to eliminate residual organic and inorganic impurities [167]. Furthermore, the sol–gel method enables precise control over particle size, porosity, and surface area [168], making it ideal for applications requiring uniform SNPs structures. However, the use of strong chemicals necessitates proper handling, neutralization, and disposal to mitigate environmental impacts.

Another promising approach for SNPs extraction is through biological processes, particularly biosilicification and biotransformation mechanisms utilizing microorganisms such as *Fusarium* species. Biosilicification is a biological process in which organisms absorb soluble, monomeric silicic acid (Si(OH)₄) and convert it into polymerized, insoluble SiO₂ [169]. Biosilicification has garnered significant research interest due to the ability of biosilicifying organisms to produce complex silica structures with remarkable morphological diversity under mild physiological conditions, such as ambient temperature, pressure,

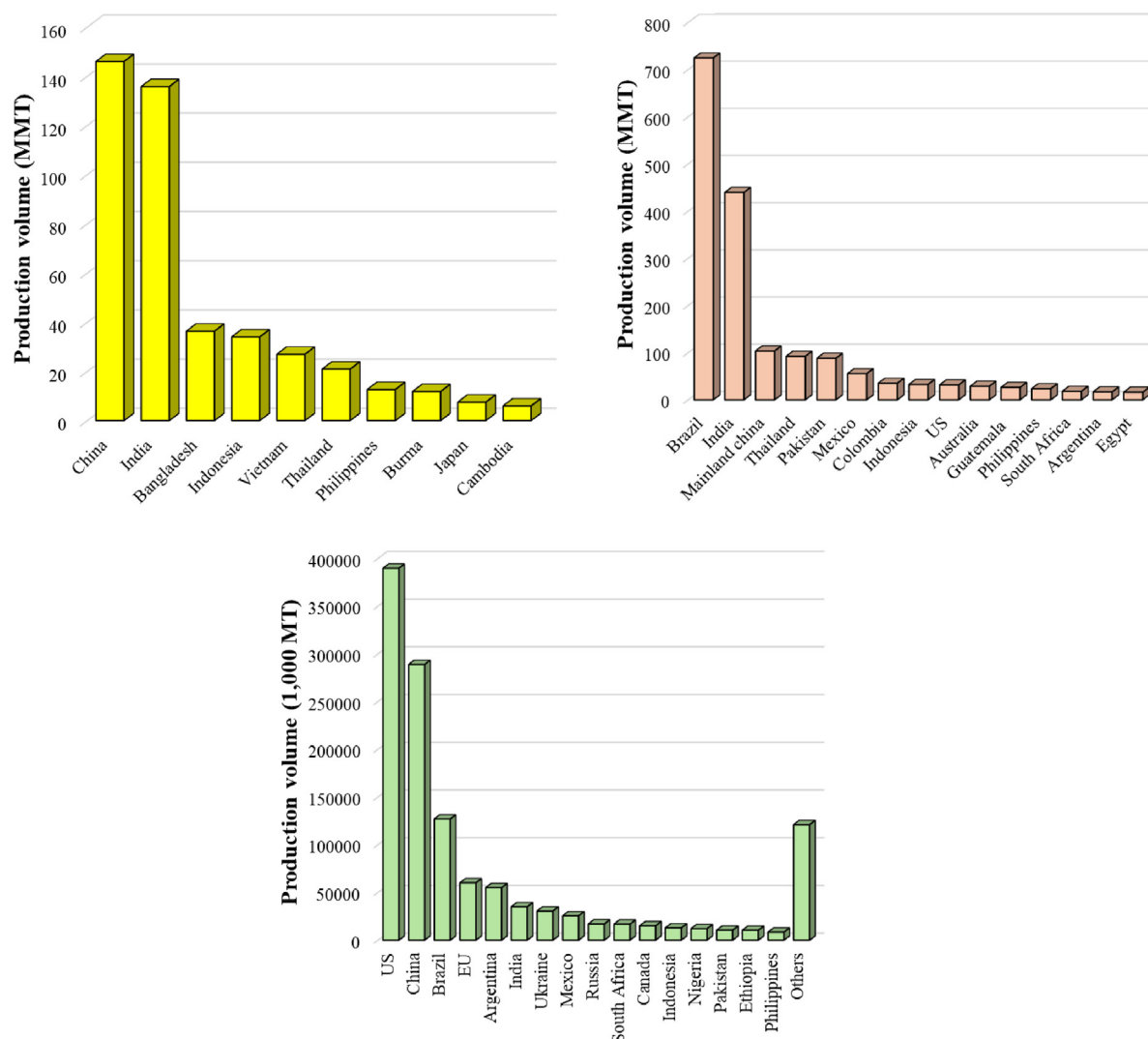


Fig. 1. Leading countries based on the production of milled rice in years 2022 to 2023 (a), sugarcane in year 2022 (b), and corn in years 2023 to 2024. Data sourced from Statista [124,125,126].

and near-neutral pH [170]. In contrast, the chemical synthesis of silica-based materials typically demands extreme conditions of temperature, pressure, and pH [169]. These methods gaining attention due to their eco-friendliness, cost-effectiveness, and ability to operate under ambient conditions. Unlike thermal and chemical methods, microbial extraction using *Fusarium* sp. offers a sustainable approach with reduced energy requirements and minimal hazardous by-products. This technique also enables precise control over particle size and morphology, making it ideal for biomedical, catalytic, and environmental applications. However, scaling up microbial processes remains challenging due to slower reaction rates and sensitivity to environmental conditions. Further studies are essential to optimize growth conditions, improve silica yield, and enhance process efficiency for large-scale applications.

4. Applications of SNPs for abiotic stress tolerance of diverse crops

Environmental stresses are significant contributors to reduced plant productivity, negatively impact plant development, and pose a major challenge to global food security and safety [171]. These stresses, which include drought, salinity, high temperatures, and heavy metal contamination, can severely limit crop growth, resulting in reduced yields and diminished agricultural sustainability [172]. With the increasing frequency and intensity of such stressors due to climate

change, it is crucial to explore innovative solutions to enhance crop resilience. A summary of recent studies on the applications of SNPs for enhancing abiotic stress tolerance in various agricultural crops is presented in Table 3. Findings from these studies have shown promising effects on improving stress tolerance across different crop types. Some highlights improved productivity and quality under saline conditions [109–111,161], drought environment [113–115,118], and improved tolerance to heavy metal contaminated environment [21,122,123]. A detailed understanding of their underlying mechanisms has been reviewed by Grewal et al. [25], Mathur and Roy [23], and readers are encouraged to refer to these articles.

Under salt stress, the excessive accumulation of salt ions in the growth environment quickly lowers the osmotic potential, leading to a physiological water deficit in plants [177]. Sodium chloride (NaCl) is a primary contributor to salt stress, where elevated levels of Na^+ compete with K^+ for uptake sites in the root system. This competition results in an accumulation of excess Na^+ within the plant, disrupting the Na^+/K^+ balance and adversely affecting plant growth and metabolism [178]. For plants experiencing salt stress, reactive oxygen species (ROS) such as hydrogen peroxide (H_2O_2), singlet oxygen ($^1\text{O}_2$), superoxide (O_2^-), and hydroxyl radicals ($\cdot\text{OH}$) are generated as a result of metabolic disturbances [179], leads to the damage of proteins, DNA, and chlorophyll [180]. SNPs alleviates salt stress in plants by regulating ion homeostasis, facilitating the uptake of beneficial potassium (K^+) ions

Table 2

Recent studies on silica nano particles (SNPs) derived from agricultural residues (focused on rice husk, sugarcane bagasse, and corn cob) since 2020.

Agricultural residue	Extraction method	SNPs properties	References
Rice husk	Precipitation method	96%–98% purity; 52.83 nm average diameter; 30–75 nm dimension range; 618 m ² /g surface area	[134]
	Leaching and co-precipitation technique	3.35 eV optical band gap; small crystallite size confirmed by XRD; UV–Vis absorption peak at 365 nm	[135]
	Leaching and co-precipitation technique	99.75% purity; 85 nm average diameter; amorphous structure confirmed by XRD	[136,137]
	Chemical method	189.64 m ² /g specific surface area; 200 nm nanoparticle size; 4.3627 mM limit of detection (LOD); 10% silica content for optimal performance; 3D interconnected structure	[138]
	Digestion by sodium hydroxide	65.95 m ² /g surface area; 50 nm average diameter; 2–15 nm pore size distribution; amorphous structure confirmed by XRD; silanol (3446 cm ⁻¹) and siloxane (1115 cm ⁻¹) functional groups identified by FTIR; spherical morphology verified by TEM and AFM; antibacterial and antifungal activity against <i>Streptococcus mutans</i> and <i>Candida albicans</i> .	[139]
	Acid refluxing and calcination	252.51 m ² /g specific surface area; 6.0278 nm mean pore radius; 180–400 nm particle size range; narrower particle size distribution compared to SiO ₂ (97–2500 nm).	[140]
	Acid leaching	99.45% purity; 21.7% yield; 335 m ² /g specific surface area; 4.95 nm average pore diameter	[84]
	Thermal decomposition	99% purity; spherical morphology; amorphous nature	[141]
	Pyrolysis-based alkaline extraction	20–170 nm average particle size; 328 m ² /g specific surface area; 98.26% purity	[142]
	Thermal decomposition	100 nm average diameter; uniform spherical shape; monodisperse distribution	[143]
Sugarcane bagasse	Acid precipitation method	45%–57% recovery efficiency; 3.96 nm particle size; 192–473.6 m ² /g surface area; amorphous structure	[144]
	L-cysteine hydrochloride monohydrate acid and tetrapropylammonium hydroxide	26–29 nm average diameter; 21.65–116.01 m ² /g surface area; crystalline cubic phases; Si–O–Si peaks at 461.23 cm ⁻¹ , 787.38 cm ⁻¹ , and 1045.99 cm ⁻¹ ; 18% loss on ignition (LOI); irregular and spherical morphology.	[145]
	Catalytic and chemical treatment process	35 nm average diameter (SiO ₂); 5.65% yield (SiO ₂); 20 nm average diameter (lignin); 10.15% yield (lignin); –35.6 mV zeta potential (SiO ₂); –13.1 mV zeta potential (lignin); 400.3 m ² /g surface area; 25.5 nm pore size; 0.624 cm ³ /g pore volume.	[146]
	Sol–gel method	90 ± 10 nm average diameter; high purity; higher specific surface area and porosity	[147]
	Sol–gel method	98.92% purity; 6–24 nm particle size range; 240 m ² /g specific surface area; 0.58 cm ³ /g pore volume; 10 nm average pore diameter.	[148]
	Sol–gel method	Primary particle size on a nanoscale; fractal structure with mesopores and macropores; lower bound water content; 60–90 °C synthesis temperature; pH 8–11; easily destructible agglomerates; potential for use in rubber compounding.	[149]
	Magnesiothermic Reduction Method	32 nm crystallite size; 73 m ² /g surface area; 0.23 cm ³ /g pore volume; 12 nm pore diameter; 446 cm ⁻¹ and 1056 cm ⁻¹ FTIR peaks (Si–O and Si–O–Si); 518 cm ⁻¹ Raman peak (crystalline silicon).	[150]
	Sol–gel method	53 nm average size; amorphous phase; siloxane and silanol groups (FTIR)	[151]
	Modified precipitation technique	97.13% purity; 85 nm average diameter; amorphous structure	[136,137]
	Sol–gel method	58.75% yield; 54–98 nm particle size range; amorphous silica	[91,92]
Corn cob	Biotransformation with the fungus <i>Fusarium culmorum</i>	47% process efficiency; 40–70 nm spherical nanoparticles	[152]
	Alkali leaching	270 m ² /g specific surface area; 12.4 nm maximum pore value; 0.678 cm ³ /g total pore volume	[153]
	Sol–gel method	108.395 m ² /g specific surface area; 2.88 nm average pore size; 569.30 m ³ pore volume; 12.75 nm crystallite size; 13.62 nm nanoparticle diameter; 0.71% porosity; 310 nm absorption wavelength; microporous characteristics.	[154]

while restricting the absorption of harmful sodium (Na⁺) ions, thus maintaining an optimal K⁺/Na⁺ balance [181,182]. Additionally, it enhances antioxidant defense systems and boosts photosynthetic efficiency under saline conditions, contributing to improved plant growth and increased tolerance to salt stress [183]. Additionally, water or drought stress in plants occurs when the availability of water falls above or below the threshold required for optimal growth and physiological

functions, often caused by prolonged periods of low soil moisture or high evapotranspiration rates. In drought conditions, water scarcity in the rhizosphere reduces groundwater potential, increases osmotic potential in plant cells, and lowers cell turgor pressure, impairing plant growth and function [184]. The effects of drought stress on plants include reduced seed germination, disruption of photosynthetic processes, and inhibition of overall plant growth [185]. Applications of

Table 3

Recent studies on the applications of SNPs for abiotic stress tolerance of diverse crops.

Abiotic stress	Plant	Results	Sources
Salt stress	Banana	Improved productivity and quality in sandy arid soil; increased yield and tolerance to salinity stress, improved sugar content of up to 50%; increased nutrient uptake, antioxidant, and osmoregulation compounds.	[109]
Salt stress	Pea plant	Improved growth, yield, and salt tolerance in <i>Pisum sativum</i> ; enhanced antioxidant defense systems, K ⁺ content, and reduced Na ⁺ accumulation under salinity stress with SNPs application.	[110]
Salt stress	Pea plant	Improved pea plant growth and yield under salinity stress; increased plant height, dry weight, flower and pod numbers, seed weight, and protein content; enhanced chlorophyll, relative water content, and antioxidant activity; reduced oxidative stress markers (MDA, H ₂ O ₂ , EL%); optimized by combined foliar application of nano silica and melatonin.	[160]
Salt stress	Wheat	Improved wheat yield and nutrient absorption under salinity stress; enhanced N, P, K, and Si contents, with reduced Na accumulation through SNPs application.	[111]
Salt stress	Grapes	Improved salt tolerance in grape seeds and seedlings under stress; enhanced growth, germination rate, and root length with combined nano-silica and Multi-Walled Carbon Nanotubes treatment: increased antioxidant enzyme activity (SOD, POD, CAT, APX) and maintained high antioxidant capacity.	[112]
Salt stress	Rice	Improved rice yield and quality under salt stress; increased yield by up to 33.3% in salt-tolerant varieties; enhanced grain-filling rate, chlorophyll content, leaf area index, and root growth; boosted antioxidant enzyme activity and reduced malondialdehyde content; improved rice processing and taste quality.	[161]
Drought stress	Hybrid maize	Foliar spraying of SNPs enhanced the growth and stress tolerance of hybrid maize.	[162]
Drought stress	Wheat	Improved drought resistance and growth in wheat; increased membrane stability, chlorophyll, carbohydrates, proteins, and phenol content; enhanced antioxidant enzyme activity and reduced oxidative stress with 100 mg/L nano-silica application, mitigating the impact of drought stress.	[114]
Drought stress	Shallots	Improved growth and physiological characteristics of shallots under water stress; enhanced chlorophyll content, leaf relative water content, and antioxidant enzyme activity (SOD) with nano-silica application at 18 g/l, particularly under optimal water conditions (80% field capacity).	[115]
Drought stress	Finger millet	Improved drought recovery, enhanced plant growth, root length, leaf area, and relative water content; increased antioxidant enzyme activity and reduced oxidative stress with 50–100 mg/L silica nanoparticle application, promoting tolerance to drought stress in drought-prone agricultural zones.	[118]
Drought stress	Maize	Improved nutrient uptake and resistance to drought stress in maize; enhanced seed concentrations of nitrogen, potassium, copper, manganese, and silica with potassium nano-silica (PNS) application; increased nutrient absorption, particularly at 100 ppm for seed nitrogen and 200 ppm for shoot potassium, manganese, and silica.	[163]
Drought stress	Rice	Improved drought tolerance in rice; enhanced leaf area, specific leaf weight, membrane stability, and chlorophyll stability with 400 ppm SNPs application; reduced proline content and improved stomatal structure, trichome length, and silicon bodies length, promoting better growth and yield under drought stress.	[116]
Drought stress	Soybean	Enhanced seed germination and seedling vigor in soybean under drought stress; improved antioxidant activity, proline content, and water retention with SiO ₂ nanoparticle priming at 500 mg/l, particularly under moisture stress conditions (−4 and −5 bars). Lower concentrations of SiO ₂ NPs optimized physiological and biochemical parameters.	[117]
Drought stress	Wheat	Improved wheat growth and drought tolerance with rice husk-extracted SNPs at 100 mg/L; enhanced shoot/root weight, relative water content, photosynthetic pigments, proline, amino acids, and antioxidant enzyme activities; reduced lipid peroxidation and soluble sugars.	[164]
Water stress	Sugar beat	Improved sugar beet growth under water stress with nano-silicon (nano-Si); enhanced chlorophyll, photosynthesis, glycine betaine, flavonols, and antioxidant enzymes at 1 mM nano-Si; reduced H ₂ O ₂ and malondialdehyde; caution with higher doses (2 mM) to avoid stress injury.	[120]
Water stress	Barley	Improved drought resistance and growth in barley; increased chlorophyll content, nutrient uptake (Fe, Zn), and yield components (1000 kernel weight, grain, and straw); reduced proline content under water stress with 100 ppm nano-silica foliar application.	[119]
Cadmium toxicity	Rice	Enhanced rice growth under cadmium stress; reduced Cd accumulation, improved photosynthetic efficiency, and antioxidant activity; increased biomass and membrane stability; downregulated Cd transporters.	[121]
Heavy metals	Pot marigold	Nano silica (1000 mg kg ^{−1}) reduced heavy metal uptake (Pb, Zn, Cu, Ni, Cr) in <i>Calendula officinalis</i> , increased soil pH by 6%, and enhanced P and K uptake by 25% and 26%, respectively. It boosted enzyme activities (urease 2.5%, dehydrogenase 23.6%, phosphatase 13.5%) and improved root and shoot growth, dry weight, and soil health.	[122]
Chromium toxicity	Rapeseed	Reduced chromium toxicity; enhanced growth, nutrient uptake, photosynthesis, and antioxidant defenses; decreased oxidative damage by 25%–36%; improved stress tolerance and detoxification mechanisms in contaminated soils.	[123]
Heavy metals + saline soil	Common bean	Enhanced growth, antioxidant defense, and nutrient uptake in <i>Phaseolus vulgaris</i> under saline and heavy metal-contaminated soils; reduced Na ⁺ , Pb, Cd, and Ni accumulation; improved photosynthesis, water status, and stress tolerance—optimal at 5 mmol/L.	[165]
Heavy metal contamination	Chameleon plant	Enhanced growth, biomass, and volatile oil composition; reduced Cd and Pb accumulation; foliar application of nano-Si was more effective and economical, reducing metal toxicity and improving plant quality.	[166]

Table 4
Benefits and drawbacks of different SNP synthesis approaches from agricultural residues in upscaling.

SNP synthesis	Benefits	Drawbacks
Thermal process	Simple method, high purity silica, suitable for large-scale production, well-established industry, minimal chemical use [156]	Required extended reaction time, potential hot spot formation, limited airflow hindering complete carbon oxidation, among other contributing factors [173,174]. Whereas, reduced in SNPs quality (surface area and amorphousness) is observed in open burning process [175].
Chemical treatment	High yield and purity, tailored for various residues, fast extraction process, established industrial process [167,176].	This method is expensive due to the relatively long reaction time (24–48 h) and the use of various chemicals and procedures [173,174]. Additionally, the use of strong acids in leaching processes has been shown to be expensive, environmentally harmful, and highly corrosive [176].
Biological process	Produced highly amorphous, likely due to the removal of impurities commonly found in biomass – which typically reduce reactivity – through leaching and microbial metabolism during digestion [176]. Biological or microbiological methods provide a cleaner and more environmentally friendly approach to SNP production.	Slow reaction rates, sensitive to environmental conditions, difficult to scale up, lower yield compared to other methods, process variability.

SNPs enhances plant water status by improving hydraulic conductivity, nutrient uptake, osmoregulation, root growth, water use efficiency, and stimulating aquaporin activity to facilitate water transport from roots to shoots under both irrigated and drought conditions [186–188]. Other than salt and water/drought stress, heavy metal contamination also offers substantial negative impacts on plants, which are primarily sourced from industrial activities, agricultural practices, and environmental contamination. The excessive uptake and accumulation of heavy metals such as lead (Pb), copper (Cu), cadmium (Cd), chromium (Cr), mercury (Hg), and others in plant tissues through the roots significantly hinder plant growth and development [189]. Numerous studies have demonstrated that SNPs are effective in promoting plant growth under heavy metal stress (see Table 3). This is accomplished by reducing the concentration of heavy metal ions, aiding in their chelation, and activating essential antioxidant systems in plants. Additionally, SNPs enhance the co-precipitation of harmful metals with silicon, facilitating their compartmentalization and regulating the expression of metal transporter genes. However, these mechanisms can vary significantly depending on plant species, genotypes, metal types, growth conditions, and the duration of stress exposure [24].

5. Challenges in SNPs scaling up

While the potential of agricultural residues, such as rice hulls, sugarcane bagasse, and corn cobs, as a reliable source of SNPs has garnered attention, the number of companies successfully producing SNPs on a large scale remains limited. For example, Guangdong Well-T Nanotech Co., Ltd. in China, is a high-tech enterprise specializing in the research, development, production, and sales of high-purity nano-silicon dioxide and polishing liquids, with an estimated production capacity of 150,000 tons [190]. Other companies such as Evonik Industries AG, Cabot corporation, NanoPore Incorporated, Fusu Chemical Co., LTD., and nanoComposix also existed [191]. However, despite the promise of these residues as sustainable feedstocks, scaling up the production of SNPs from agricultural waste faces several significant challenges. Environmentally, thermal and chemical extraction methods often lead to high energy consumption, greenhouse gas emissions, and hazardous by-products, posing risks to ecosystems and human health. Although biological processes offer eco-friendlier alternatives, they are hindered by slow reaction rates, sensitivity to environmental variations, and difficulties in scaling up production. Additionally, variability in feedstock composition due to differences in crop species, geographical location, and soil conditions can affect silica yield and purity, complicating process standardization and reproducibility. A summary of the benefits and drawbacks of these three synthesis methods, which could potentially challenge SNPs production at a commercial scale,

is depicted in Table 4. Addressing these environmental challenges requires advancements in extraction technologies to minimize emissions, enhance resource efficiency, and maintain consistency in silica quality.

6. Sustainability triangle and future trends

It is evident that SNPs derived from agricultural residues have not only excelled in engineering and interdisciplinary applications but also shown promising results in mitigating the adverse effects of diverse abiotic stresses in plants. However, despite these promising outcomes, this technology faces challenges, particularly in scaling up production, as discussed in previous section of this article. While using agricultural residues as raw materials for nanomaterial synthesis aligns with sustainable practices (such as SDG 12: Responsible Consumption and Production), it is crucial to recognize the sustainability triangle, which highlights the interconnection between economic, environmental, and social sustainability (see Fig. 2) [192–194]. To produce sustainable nanomaterials from agricultural residues, the three pillars of sustainability – environmental, social, and economic – must work in harmony [83,194]. As emphasized by [195], assessing a new biorefinery process requires an integrated focus on both sustainability and technical feasibility, from the early design stages through to full development, to ensure long-term effectiveness and production capacity. Before scaling up biorefineries for nano-silica, these sustainability dimensions must be thoroughly evaluated to guarantee viability and efficiency. Economic sustainability ensures that the process remains cost-effective, scalable, and profitable over time; environmental sustainability emphasizes waste minimization, carbon emission reduction, and efficient resource use through circular economy practices [194]; and social sustainability highlights the importance of equitable benefits, such as job creation, rural development, and the adoption of eco-friendly technologies to enhance societal well-being.

In the following subsections, recent advancements in SNPs production will be examined through the lens of these sustainability pillars. Emphasis will be placed on how current studies address economic, environmental, and social considerations. Additionally, emerging research gaps and potential future directions will be highlighted to guide further innovation and development in sustainable SNPs biorefineries.

6.1. Process optimization

While generalized methods for SNPs synthesis from various agricultural residues are already available in the literature, it is crucial to develop an optimized procedure for SNPs synthesis before scaling up. Process optimization focuses on eliminating inefficiencies during synthesis, maximizing yield and quality while minimizing operating

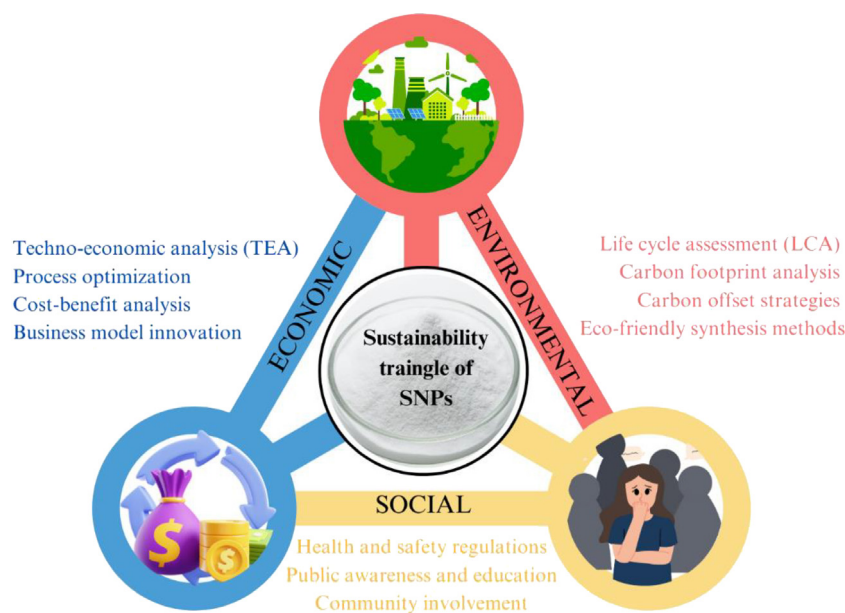


Fig. 2. Sustainability triangle of SNPs based on future perspectives.

costs [83,196]. The optimization approach aims to establish the foundation for the techno-economic analysis [194]. Techniques such as response surface methodology (RSM) and design of experiments (DOE) are widely used to evaluate and optimize critical process parameters systematically [197]. This approach is an essential step prior to assessing the three pillars of sustainability presented in Fig. 2. In recent years, various studies have been conducted to optimize process parameters for the synthesis of SNPs, focusing on different responses such as yield, particle size, polydispersity index, and crystallinity index (see Table 5). For instance, Gun et al. [198] optimized the silica extraction process from rice husk using RSM, considering temperature, potassium hydroxide-to-rice husk ratio (KOH:RH), and exposure time as the process variables, with yield as the response. Similarly, bagasse ash, recognized for its high silica content, has been investigated for SNP production using the hydrothermal method. Optimization using RSM with a Central Composite Design (CCD) predicted an optimal synthesis temperature of 152.67 °C and a reaction time of 6 h, resulting in particles with a size of 276.29 nm and a polydispersity index (PDI) of 0.1896. Experimental validation yielded particles with an average size of 330.39 nm and a PDI of 0.3580, indicating only a slight deviation from the predicted values [199].

Another technique employed for the optimization of SNP synthesis is the Taguchi method, a robust design approach that utilizes orthogonal arrays to systematically study the influence of multiple process parameters with a reduced number of experimental trials [204,206,207]. Compared to RSM, which is based on regression models and typically requires more experimental runs, the Taguchi method is more efficient in identifying optimal conditions under constrained resources. Moreover, it emphasizes the minimization of variability and the enhancement of process robustness by analyzing signal-to-noise (S/N) ratios [208–211], making it particularly advantageous in handling noise factors that can affect nanoparticle synthesis. This technique has been successfully applied in the processing of bamboo leaves as a silica source, achieving a silica purity of 94.6% and producing nanoparticles with sizes ranging from 33 to 45 nm [203]. Additionally, process optimization on SNPs synthesis from rice husk ash was conducted using the Taguchi method, achieving 98.26% silica purity with 4N NaOH, 20 ml/g alkali volume, and 60 min of treatment. Alvarado et al. [93] previously reported that Artificial Neural Networks (ANN) address certain limitations of RSM, particularly its constraint to local analysis. Specifically, the response surface generated through RSM is

only valid within the defined experimental range, limiting its applicability to conditions outside the specified factor levels. This implies that RSM may yield inaccurate predictions when extrapolating to system parameters beyond the scope of the original study. ANN modeling also validated the process, demonstrating its feasibility and efficiency for sustainable SNPs production [212]. Other than agricultural residues, process optimization on SNPs extraction from pumice [213], perlite [214], tetraethyl orthosilicate (TEOS) [215], and dune sand via sol-gel method [216] also existed.

Despite the available data, there is a clear need for further process optimization for SNPs synthesis from a variety of agricultural residues, incorporating different synthesis procedures (considering responses in Fig. 3) and optimization techniques with enhanced predictive capacity. It is important to note that the SNPs extraction process is delicate and highly influenced by the type and composition of raw materials, with variability arising from factors such as geographic location, cultivation practices, and crop variety. These factors can significantly impact the efficiency and quality of the synthesized SNPs, highlighting the importance of considering such variables in the optimization process.

6.2. Techno-economic analysis

The growing interest in sustainable materials derived from agricultural residues underscores the importance of performing a techno-economic analysis (TEA) to evaluate the viability of these emerging technologies. As novel processes are developed to convert waste biomass into high-value nanomaterials, such as SNPs, it is not enough to focus solely on technical feasibility. Although upgrading high-volume, undervalued agricultural residues often leads to reduced pollutant emissions, economic viability is not always guaranteed in this context [195,217]. Economic feasibility must also be considered to assess the commercial potential and long-term sustainability of these technologies. TEA serves as a powerful decision-making tool, guiding researchers, investors, and policymakers in determining whether a technology can transition from the laboratory to full-scale production and market implementation. In the context of a circular bioeconomy, TEA aligns innovations with economic and environmental goals, ensuring that resource-efficient solutions can also be economically competitive.

Techno-economic analysis refers to a comprehensive evaluation that integrates process engineering and economic modeling to assess the cost-effectiveness of a production system [218–220]. Typically,

Table 5

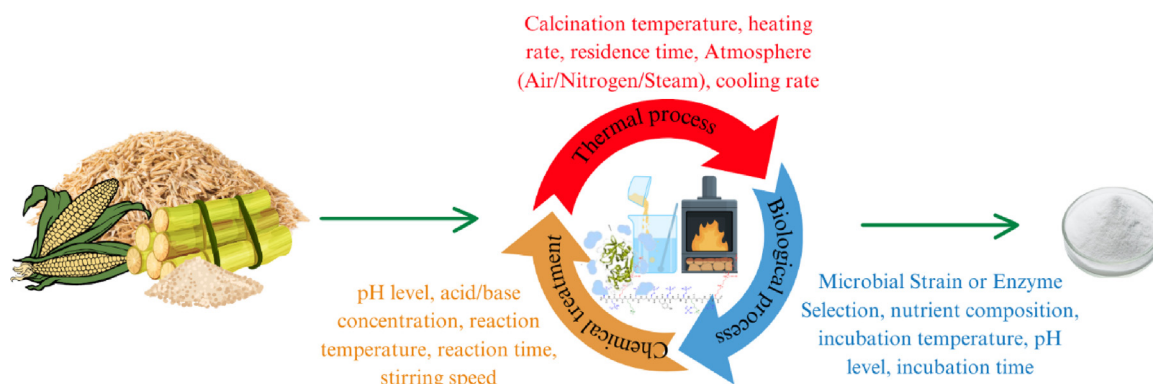
Recent studies on process optimization of Silica Nanoparticle synthesis from agricultural residues and diverse conventional sources.

Raw material	Optimization technique	Parameters	Optimized process parameters	References
Rice husk ^a	L ₁₆ Taguchi OA, ANN	Factors: NaOH concentration, alkali volume, reaction time Response: yield	NaOH concentration (4 N), alkali volume (20 mL/g), reaction time (60 mins) with 98.26% yield.	(Pathak et al. 2020a)
Bagasse ash ^a	RSM CCD	Factors: Reaction time, temperature Response: particle size, polydispersity index	Temperature (152.67 °C) and reaction time of 6 h results to an average particle size of 276.288 nm and PDI value of 0.189642	[200]
Rice husk ^a	MOGA, BPANN, RSM	Factors: leaching temperature (65–85 °C) & time (0.5–1.5 h) and calcination temperature (450–650 °C) & time (1–3 h) Responses: crystallinity, RHA recovery, volume of % silica	Leaching temperature (75 °C), leaching time (1 h), calcination temperature (550 °C) and calcination time of 2 h. The optimized model was validated with observed results and found to be a well-fitted with absolute errors of 7.42%, 2.03%, and 4.26% for % crystallinity, RHA recovery, and vol% of silica, respectively.	[201]
Teff straw ^a	RSM CCD	Factors: temperature, NaOH concentration, rotational speed, and extraction time Response: yield, density	temperature (94.98 °C), NaOH concentration (1.5 M), rotational speed (499.57 RPM), and extraction time (99.99 min) with the yield of 85.85% and	[202]
Bamboo leaves ^a	Taguchi OA	Factors: calcination time, temperature and sodium hydroxide concentration Response: particle size	Time (4 h), temperature (700 °C), and NaOH concentration (4 N) with 94.6% purity and 33 nm to 45 nm particle size.	[203]
Expanded perlite ^b	3-level L ₉ Taguchi OA	Factors: acid concentration, alkaline concentration, alkaline temperature and pH Response: particle size	Acid concentration (4 M), alkaline concentration (2.5 M), alkaline temperature (80 °C) and pH (10) with 99% purity and average particle size ranges of 8 nm to 74 nm.	[204]
Pumice ^b	RSM CCD	Factors: reaction temperature, reaction time and molar ratio ($n = n_{NaOH}/n_{Silica}$) Response: Yield	Temperature (100 °C), reaction time (6 h), and molar ratio (17.5) with the yield of 95.83%	[205]

RSM – response surface methodology; CCD – central composite design; OA -orthogonal array; ANN – artificial neural network; MOGA - multi-objective genetic algorithm; BPANN - back-propagation artificial neural networks.

^a Agricultural residues and other non-conventional sources.

^b Conventional sources.

**Fig. 3.** Process variables for SNPs synthesis via thermal, chemical and biological processes.

TEA involves several key indicators such as total capital investment (TCI), operating costs (OPEX), net present value (NPV), internal rate of return (IRR), payback period (PBP), and minimum selling price (MSP) [221–224]. These metrics help identify cost drivers, determine breakeven points, and compare different process configurations. Sensitivity and uncertainty analyses are also crucial in TEA, providing insight into how variable parameters – such as raw material costs, energy use, or product yield – affect overall economic performance. By quantifying these indicators, TEA supports evidence-based optimization and strategic planning in technology development.

In recent years, TEA has been increasingly applied to a wide range of nanomaterials synthesized from agricultural residues and other sources, such as nano-silica [225–227], nano-chitin [228], nanocellulose [229–231] nano-zeolites [232], and other nanoparticles [233–236]. For the production of bulk silica particles from serpentine mineral, Çakan and Gönen [237] conducted a TEA under optimized process parameters: a temperature of 67.3 °C, acid concentration of

4.8M, and reaction time of 57.1 min. The TEA was carried out for various annual plant capacities ranging from 4000 to 16,000 tons. The analysis identified 12,000 tons per year as the optimal capacity, offering the lowest production cost. At this scale, the total capital investment (TCI) was estimated at \$40.25 million, while the operating cost was approximately \$4.24 per kilogram of silica. The facility's payback period was determined to be 4.8 years. For nanosilica, Kim and Lim [226] conducted a TEA on the production of electricity and nano-porous silica from 50 tons per day (t/d) of rice husk. The electricity generation process from rice husk exhibited a net efficiency of 15%. Using the same input amount of RH (approximately 2 tons per hour), the process yielded 278 kg/h of NPS. However, electricity production alone was found to be unprofitable due to a negative return on investment (ROI). In contrast, the production of NPS from RH showed more favorable economics, with a total capital investment (TCI) of \$13.7 million and an annual total production cost (TPC) of \$1.5

million. The estimated ROI for NPS production was 3.7% per year, with a projected payback period (PBP) of 17.5 years.

Despite the growing number of studies on nano-silica production, TEA applications in this field remain limited and often narrowly focused. While generalized synthesis methods are commonly applied across various agricultural residues, differences in feedstock composition – such as silica content, ash characteristics, and impurity levels – can significantly influence both process efficiency and economic performance. These variations highlight the need for more feedstock-specific TEAs that account for such disparities. Moreover, existing TEAs typically evaluate fixed-scale operations, overlooking the potential economic implications of different production capacities and regional contexts. There is also a need to conduct thorough sensitivity analyses to identify key cost drivers – such as energy consumption, chemical usage, and capital investment – which are essential for process optimization and informed scale-up. Expanding TEA efforts in nano-silica production, particularly from diverse agricultural wastes, will provide crucial insights for guiding industrial adoption and ensuring economic feasibility.

6.3. Life-cycle assessment

In the context of modern sustainable development, environmental sustainability has become a fundamental pillar alongside economic and social sustainability—together forming the so-called sustainability triangle [193,238,239]. Environmental sustainability ensures that natural ecosystems are preserved and that human activities do not exceed planetary boundaries [239]. In materials science and nanotechnology, this dimension is particularly important due to the growing concerns about emissions, energy use, and waste generation. Although the drive for innovative nanomaterials like silica nanoparticles is rooted in performance enhancement across various industries, including agriculture, it is critical that these innovations align with the principles of sustainable production and circular economy. Ensuring the environmental feasibility of SNP production – particularly from agricultural residues – can offer a unique opportunity to valorize waste while minimizing ecological footprint.

To comprehensively assess the environmental impacts of a biorefinery, researchers and policymakers increasingly rely on scientific tools and frameworks that offer quantifiable and comparable metrics, such as energy analysis, probabilistic modeling, energy evaluation, Life Cycle Assessment (LCA), carbon footprint, and water footprint [240–245]. Among the various approaches, LCA has emerged as a gold-standard methodology for evaluating the environmental burdens associated with a product or process throughout its entire life cycle (cradle to grave or cradle to gate) [68,69,246]. LCA covers all stages from raw material extraction, production, use, and end-of-life disposal or recycling [247]. LCA is defined as a “cradle-to-grave” approach used to assess the environmental impacts throughout a product’s entire life cycle [219,248,249]. In the case of nano-silica derived from agricultural residues, LCA can play a critical role in identifying emission hotspots, evaluating resource efficiency, and comparing the sustainability of different synthesis routes such as sol-gel processing, combustion, or acid-leaching techniques.

LCA studies are conducted in alignment with the guidelines and standards established by the International Organization for Standardization (ISO), including ISO 14040, ISO 14041, ISO 14042, and ISO 14043 [247]. The LCA methodology follows an iterative process comprising four closely interconnected stages: goal and scope definition, life cycle inventory analysis, life cycle impact assessment, and interpretation of results [250], (see Fig. 4 for LCA framework). What makes LCA especially attractive is its comprehensive and holistic nature, allowing for a cradle-to-grave or cradle-to-cradle analysis [251,252]. Unlike traditional carbon accounting or energy audits, LCA provides a multi-dimensional view of environmental impacts, including carbon footprint, water use, ecotoxicity, eutrophication, and resource

depletion. This makes it an essential decision-support tool in green technology development and policy planning.

Recent studies have demonstrated LCA’s relevance in quantifying greenhouse gas (GHG) emissions, energy consumption, and other environmental metrics associated with the production of diverse nanomaterials [251,253–256]. In nano-silica production, LCA is particularly useful for identifying hotspots in energy consumption and emissions throughout the process, enabling targeted improvements to enhance environmental performance. For bulk silica particles derived from RHA, LCA results revealed that the overall climate change (CC) impact of the extraction process is approximately 7.26 kg CO₂ equivalent per kilogram of silica produced [257]. According to their study, the greater climate impact of RHS production is primarily due to the use of strong acids and bases during chemical treatment, as well as the high consumption of electrical energy. Gu et al. [258] demonstrated that the environmental footprint of SNP production is significantly affected by the source of electricity – such as wind, hydropower, solar thermal, and photovoltaic energy – with green energy alternatives reducing emissions by 40%–90%. Similarly, Gu et al. [259] conducted a comprehensive LCA on rice husk-derived nano-silica, revealing that the choice of chemicals used in extraction and precipitation significantly influences environmental emissions. Their study showed that using NaOH and HCl resulted in lower emissions compared to Na₂CO₃, H₂SO₄, and HNO₃, with further emission reductions of 20%–60% achieved by switching to cleaner electricity sources like wind, solar, and hydropower. These findings highlight the importance of selecting both greener reagents and renewable energy sources in nano-silica production. Another eco-friendly approaches, such as oxalic acid pre-treatment and controlled calcination, have shown promise in enhancing both environmental and material performance. For example, in the study of Dominic et al. [127], SNPs produced from millet husk through mild oxalic acid pre-treatment and calcination at 700 °C for 2 h yielded particles of 10–20 nm with higher purity, greater surface area (238 m²/g), and improved thermal resilience compared to untreated samples. LCA analysis revealed that this method reduced global warming potential by 40%, human carcinogenic toxicity by 38%, and terrestrial acidification by 38% compared to rice husk-derived SNPs. LCA on SNPs derived from SCBA was also conducted by Mamidi et al. [260] where the highest global warming potential (GWP) was noted in SNPs produced with HNO₃ followed by HCl and H₂SO₄ with a CO₂ equivalent of 12.59 kg, 12.40 kg, and 7.51 kg, respectively.

Recent studies present conflicting views on the sustainability of nano-silica from agricultural residues—while feedstock reuse reduces costs and emissions, high energy inputs during processing may offset these gains [261]. Inconsistencies in datasets, assumptions, and system boundaries further challenge the comparability of LCA results [262,263]. This highlights the need for standardized, nano-specific LCA frameworks that consider unique properties and full life cycle stages. Future research should integrate TEA, dynamic models, and open-access data to improve accuracy, transparency, and decision-making.

6.4. Social sustainability

Social sustainability is a critical component of sustainable development, complementing the environmental and economic dimensions to form the three interconnected pillars of sustainability [264]. While the environmental and economic aspects of nano-silica production from agricultural residues have received considerable attention, the social dimension remains comparatively underexplored. Social sustainability emphasizes the promotion of human well-being, equity, labor rights, and community development throughout the life cycle of a product or process. In the context of nano-silica derived from agricultural waste, it entails the consideration of occupational health and safety, equitable distribution of socio-economic benefits, community participation, and the long-term viability of rural livelihoods.

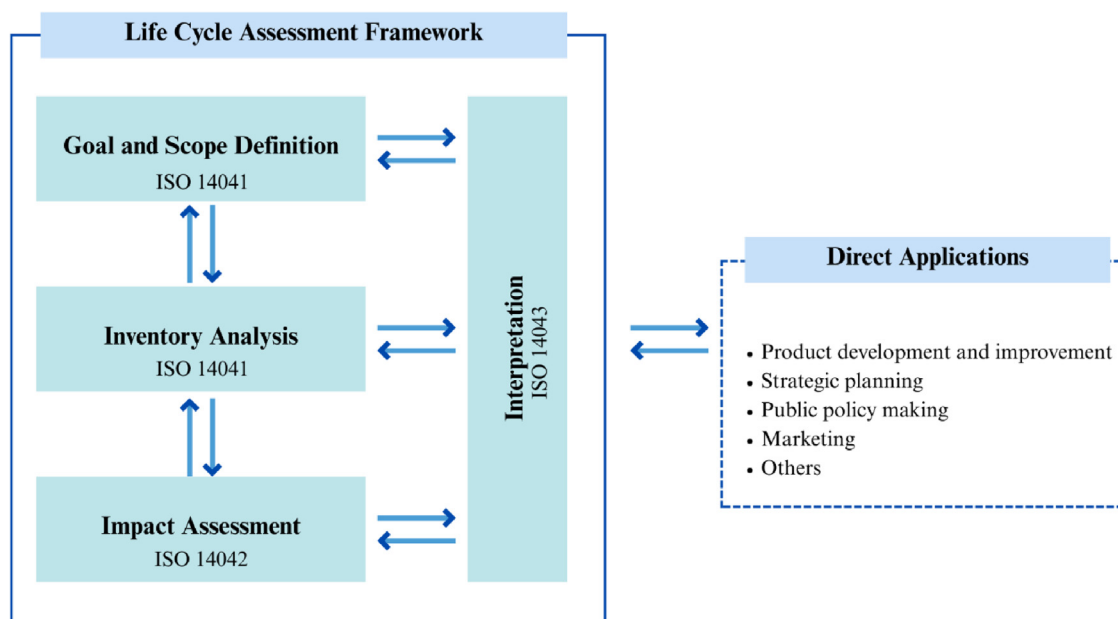


Fig. 4. The framework of LCA consists of four interrelated phases, as defined by the ISO 14040 standard.

In terms of acceptability, Jacobs et al. [265] emphasized that understanding the risks associated with nanotechnology and the use of nano-silica is essential for ensuring both social acceptance and the safe application of these technologies. Their probabilistic risk analysis of nano-silica in food products indicated, with 95% confidence, that at least 99% of the Dutch population would be exposed to no significant health risk. They also noted that a major challenge in assessing nanoparticle risks is the lack of sufficient data, which leads to considerable uncertainty in the evaluation process. For health-related concerns, recent reports have identified amorphous silica nanoparticles as potent inducers of inflammation, cytotoxicity, and altered immune responses in humans [266]. These findings underscore the need for further research to assess the long-term effects of exposure to such nanoparticles, particularly in food products and consumer goods, to ensure public safety and inform regulatory frameworks. Unlike economic and environmental sustainability, which can be quantitatively assessed using well-established tools such as TEA and LCA, social sustainability is typically evaluated using qualitative methods [195]. It has also been emphasized that a significant degree of subjective interpretation is often involved when evaluating the pillars of sustainability—particularly in the case of the social dimension. This subjectivity arises from the lack of standardized indicators and universally accepted metrics, making it challenging to consistently assess social impacts across different contexts and projects.

6.5. Future directions

The future application of SNPs in plant stress mitigation is expected to be increasingly guided by integrative approaches across the sustainability triangle—encompassing environmental, economic, and social dimensions. While recent progress has focused on process optimization using agricultural residues, future research must improve predictive models that account for feedstock variability influenced by geographic and agronomic factors. Enhanced optimization techniques and synthesis protocols tailored to specific biomass sources will be essential to ensure consistent SNP quality and yield, addressing the complex interactions between raw material composition and process efficiency.

Economically, there is a growing need for feedstock-specific and scale-sensitive techno-economic analyses (TEAs) that reflect real-world variability in silica content, energy consumption, and capital require-

ments. Current TEAs often apply generalized assumptions and fixed-scale evaluations, overlooking the implications of different production capacities and regional contexts. Future directions should prioritize dynamic TEA models coupled with life cycle assessments (LCAs) to offer a more comprehensive understanding of sustainability trade-offs. Standardized, nano-specific LCA frameworks are also critical to resolve existing inconsistencies in data and system boundaries, thereby enabling more reliable comparisons and guiding sustainable industrial deployment.

Social sustainability, often assessed qualitatively, requires greater standardization through objective and context-specific indicators to evaluate factors such as labor conditions, community impact, and technology acceptance. Addressing this gap is crucial for inclusive technology adoption, particularly in rural settings where agricultural residues are abundant. At the same time, expanding the agronomic application of SNPs, particularly for mitigating abiotic stresses like drought and salinity, will be key to climate-resilient agriculture. Future studies should focus on long-term field trials, dose optimization, and environmental safety to ensure that the benefits of SNPs are aligned with broader sustainability goals.

7. Conclusions

In conclusion, this review assessed the potential of agricultural residues such as rice husks/hulls, sugarcane bagasse, and corn cobs as sustainable sources for silica nanoparticles (SNPs) production, emphasizing their abundance and richness in silica. The review also explored three major synthesis approaches – thermal processes, chemical treatments, and biological methods – each offering distinct advantages and challenges. Despite the promise of these residues as feedstocks, scaling up SNPs production faces several significant hurdles. Thermal and chemical extraction methods are energy-intensive, often resulting in high greenhouse gas emissions and hazardous by-products that pose risks to both ecosystems and human health. While biological processes present eco-friendly alternatives, they are constrained by slow reaction rates, environmental sensitivity, and scalability challenges. Furthermore, variability in feedstock composition, influenced by factors such as crop species, geographical location, and soil conditions, can impact silica yield and purity, complicating standardization and reproducibility of the process. Considering the three pillars of sustainability, process optimization is a vital step prior to assessing economic,

environmental, and social sustainability. Despite the global call for sustainability, studies regarding TEA, LCA, and social sustainability are still lacking, which suggests that future studies should prioritize these areas. Comprehensive evaluations of these aspects will provide a more holistic understanding of the viability and scalability of SNPs production, ensuring that the technology can be both economically feasible and socially responsible while minimizing environmental impact.

CRedit authorship contribution statement

Arsenio D. Bulfa: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Conceptualization. **Marjun C. Alvarado:** Writing – review & editing, Writing – original draft, Conceptualization. **Pearl B. Sanchez:** Supervision. **Ma. Lourdes S. Edaño:** Validation, Supervision. **Pompe C. Sta. Cruz:** Validation, Supervision. **Eureka Teresa M. Ocampo:** Writing – review & editing, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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